

Response to Reviewers

Editor comments

The manuscript presents a useful applied comparison of Bayesian and frequentist meta-analysis, but several claims about the relative merits of the two frameworks are overstated or lack sufficient support. The authors should refocus the paper as a comparison of two specific implementations rather than broad frameworks, clarify technical inaccuracies, and provide supporting analyses for statements about robustness to outliers and priors. Both reviewers have provided highly constructive and meaningful suggestions that should be carefully addressed by the authors. - Rongji Mu

Response to the Editor

We would like to thank the editor and reviewers for their careful reading of our manuscript and their valuable feedback. Below we respond to each point in turn; revised manuscript text is quoted verbatim.

Reviewer 1

This manuscript presents a systematic meta-analysis of the association between lead exposure and children's IQ and compares frequentist and Bayesian meta-analytic frameworks. The study is well organized and the methodology is clearly described. However, some statements comparing Bayesian and frequentist approaches appear overly strong or technically imprecise. A broad comparison between Bayesian and frequentist philosophies involves complex statistical issues that require extensive methodological expertise. This manuscript would be stronger if it focused on clearly presenting how the two approaches differ in this specific meta-analysis and how these differences affect the interpretation of the results, rather than making general claims about the superiority or limitations of one framework over the other. The following comments may help improve the manuscript.

Major comments

Reviewer 1, comment 1

In the Abstract, the authors state: "Sensitivity analyses showed robustness to RoB and prior specification." This statement may be misleading. For the prior sensitivity analysis, the results only indicate that, in this specific dataset, the likelihood information dominates the influence of the selected priors. This should not be interpreted as a general property of Bayesian analysis. Prior specification can be important, especially in small meta-analyses with limited evidence. In addition, the statement regarding robustness to RoB requires clarification. According to Table 5, the frequentist estimates appear at least similarly robust after excluding studies with high or moderate RoB. Therefore, the Abstract should avoid implying a particular advantage of the Bayesian framework.

Response

Thank you for this thoughtful comment. We agree and have revised the abstract to clarify that both frameworks were similarly robust to exclusion of studies with high or moderate RoB, and that Bayesian estimates were insensitive to the alternative priors examined in our sensitivity analysis — which is not a general property of Bayesian analysis.

Revised: Abstract – Results

“Sensitivity analyses showed both frameworks were robust to RoB, and Bayesian estimates were insensitive to tested priors.”

Reviewer 1, comment 2

The manuscript states: “The Bayesian approach was less sensitive to outlier studies reporting substantial uncertainty.” However, no specific analysis is provided to support this statement. The authors should clarify which studies are considered outliers and quantify how much the pooled estimates change after excluding these studies.

Response

Revised

Reviewer 1, comment 3

In the Background, the authors state: “Unlike frequentist confidence intervals, which are often misinterpreted as probabilistic...” This statement requires clarification. The issue is not that frequentist confidence intervals lack a probabilistic interpretation. Under the frequentist framework, confidence intervals are random intervals generated by a repeated sampling procedure and have a specified coverage probability. The common misunderstanding is that, after observing the data, the calculated confidence interval can be interpreted as having a 95% probability of containing the fixed but unknown parameter. Therefore, the distinction between frequentist confidence intervals and Bayesian credible intervals should be explained more precisely. The current wording may unintentionally suggest that confidence intervals are non-probabilistic, which is not accurate.

Response

Revised

Reviewer 1, comment 4

In Section 3.3, the manuscript reports that the Bayesian model gives a larger between-study variance (τ^2) but a slightly smaller I^2 compared with the frequentist model. The authors state in Section 2.4.1 that I^2 was calculated using the typical within-study variance from the

frequentist framework. However, the exact calculation of Bayesian I^2 is unclear. Please provide the formula or detailed explanation for how Bayesian I^2 was obtained and explain why a larger estimate of τ^2 resulted in a smaller I^2 .

Response

Revised

Reviewer 1, comment 5

In Section 3.3, the authors state: “In the context of meta-analysis, shrinkage is comparable to the inverse-variance weighting in the frequentist framework.” This statement is potentially misleading. Bayesian shrinkage of study-specific effects occurs because posterior estimates incorporate information from the hierarchical distribution and borrow information from the overall mean. In contrast, inverse-variance weighting in frequentist meta-analysis determines the contribution of each study to the pooled effect estimate but does not change the individual study estimate. The authors should clarify whether they are discussing the estimation of individual study effects or the weighting mechanism used to obtain the pooled effect.

Response

Revised

Reviewer 1, comment 6

In Section 3.4, the authors conclude: “This shows that the results are robust to prior specification.” This conclusion should be restricted to the specific prior distributions examined in this study. The results demonstrate robustness under the selected scenarios, but do not imply that Bayesian results are generally insensitive to prior choices.

Response

Revised

Reviewer 1, comment 7

In the Methods section and Discussion, the authors state that all code and data are publicly available on GitHub. However, the provided GitHub link is currently inaccessible. Since reproducibility is an important component of this study, the authors should provide a valid and accessible repository link before publication.

Response

Revised

Reviewer 1, comment 8

In the Discussion, the statement that the Bayesian model yielded lower pooled effects “due to posterior shrinkage” is incomplete. The difference between Bayesian and frequentist estimates also reflects the incorporation of prior information and should not be attributed solely to shrinkage.

Response

Revised

Minor comments

Reviewer 1, comment 9

In Supplementary File 5, Figures 5a and 5b require clearer descriptions. The figure legends should specify what is displayed and how the prior predictive and posterior predictive checks were conducted.

Response

Revised

Reviewer 1, comment 10

In Section 3.4, the manuscript states that sensitivity analysis results are presented in Table 4, but they appear to be reported in Table 5. Please correct this reference.

Response

Revised

Reviewer 2

This manuscript compares Bayesian and frequentist random-effects meta-analysis using the association between blood lead levels and children’s IQ as a case study. The topic is relevant, and the use of the same studies and effect estimates in both analyses makes the comparison easy to follow. However, several methodological and reporting issues should be addressed before publication

Major comments**Reviewer 2, comment 1**

The manuscript compares one REML model with one Bayesian normal-normal hierarchical model. This is a useful applied example, but it does not support broad conclusions about Bayesian versus frequentist frameworks. With only 12 studies and substantial heterogeneity, the frequentist analysis could also include a small-sample adjustment. The authors should either broaden the comparison or describe the study more cautiously as a comparison of two specific implementations.

Response**Revised**

Reviewer 2, comment 2

Although the coefficients were converted to the same logarithmic scale, the included studies differ considerably in age, exposure range, study design, IQ instrument, and covariate adjustment. Some estimates also come from models adjusted for other environmental exposures. Therefore, the coefficients may not represent exactly the same estimand. Authors need more caution explaining those numbers.

Response**Revised**

Reviewer 2, comment 3

The manuscript states that study estimates in both panels of Figure 2 are shrunk toward the pooled estimate. However, the Bayesian panel appears to show posterior study-specific estimates, whereas the frequentist panel shows the original study estimates. Inverse-variance weighting of the pooled mean is not the same as shrinkage of study-specific effects. The statement that frequentist methods implicitly use flat priors should also be removed or qualified. Similarly, the claim that the Bayesian model is less sensitive to outliers is too strong because the model still assumes normally distributed random effects and no formal outlier analysis is presented.

Response**Revised****Minor comments**

Reviewer 2, comment 4

Different values appear in the Abstract, Results, Figure 2, and Table 4. Check the consistency.

Response

Revised

Reviewer 2, comment 5

The text refers to children aged 0–11 years, while Table 1 includes participants under 18, pregnant women, prenatal exposure, and methylmercury.

Response

Revised

Reviewer 2, comment 6

Check the confidence interval for frequentist heterogeneity estimate. For examples, the reported value tau2 is 3.3 with 95% CI [1, 3.3]. Seems not usual that the point estimate is equal to the upper limit. Need to double check this.

Response

Revised